M3 Challenge 2025:

Team #17470

March 1st, 2025

Executive Summary

Dear Birmingham city council,

This report aims to provide you with the mathematical models and plots to inform you in decisions dealing with two topical relevant issues - bouts of extreme heat (commonly referred to as heatwaves) and their effects on the temperature inside of housing units, and the effects of these heatwaves on the electrical infrastructure and power draw of Birmingham and its citizens.

This report is broken down into three main parts - modelling the temperature change inside a house compared to the outside temperature during a period of extreme heat; modelling the changes in energy draw during a period of extreme heat (attributed to the increase of use of cooling appliances such as air-conditioning systems, fans and the like); and creating a metric for ranking constituencies in order of vulnerability during heat waves.

Our findings show that certain designs of houses, like low rise 1 bedroom apartments, are much more resistant to heat during a heatwave than other residential buildings like flats. Furthermore, despite a significant expected increase in temperature over the next 20 years, we are expecting improvements in cooling technology to outpace the warming, in fact decreasing the cities peak power consumption during hot weather due to higher efficiency.

We have also found that certain constituencies - notably Birmingham Hall Green and Moseley - have a higher vulnerability to heat waves due to an older population and less resources within the area, along with more crowding in houses.

We have used our findings to draft some recommendations for managing this, and are provided below.

- Construct new buildings with green roofs/walls [1], and attempt to introduce green roofs to already existing buildings where possible.

  Justification: Because of an effect known as the "heat island effect" [2], urban spaces typically get significantly hotter than surrounding green areas. Using green roofs significantly reduces this, in some cases reducing the temperature of nearby air by up to 10 $^{\circ}C$ [3].

- When reviewing planning requests, prioritise approval for low-rise apartment blocks in areas with high overcrowding.

  Justification: The models used in Questions 1 and 3 have shown that low rise apartment buildings and bungalows are the most efficient for resisting rising temperatures, they also spread people out to reduce overcrowding and allow for more green spaces as it spreads the city wider.

- Prepare for increasing temperatures over the next 20 years, outfitting more houses to be able to cope with the higher temperatures we can expect.

  Justification: We have predicted temperatures are likely to rise over the next 20 years and with many vulnerable groups the city needs to be able to cope with more frequent and severe heatwaves.

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1 Question 1: Hot to Go

1.1 Restatement of the Problem

In this problem, we are tasked with creating a model to predict the indoor temperature of non-air-conditioned dwellings during a heat wave over a 24 hour period in Birmingham. We then had to test our model against the data provided.

1.2 Assumptions

- The temperature inside the house is uniform.

  Justification: Differences between rooms will be negligible and largely dependant on the internals walls of a home which can't be generalised in our model.

- Each housing unit is a cuboid with a square base.

  Justification: Houses tend to be roughly square-based and alternate cuboid shapes would complicate the model without making them noticeably more accurate.

- Nothing other than people in a building are producing heat.

  Justification: People won't turn on heating during a heatwave.

- The soil temperature is $12°C$.

  Justification: [4], taking maximum value because we are looking at heatwaves.

- There are no cooling methods in place such as open windows or air conditioning.

  Justification: Modelling energy loss due to air flow throughout a building would be near impossible and depend heavily on building structure, which was unavailable to us.

- The only factors effecting heat energy in a house are people in the house, outside temperature and sunlight.

  Justification: These are the main factors we identified, and including more would further complicate the model for comparatively little benefit to our model.

1.3 Model Development

1.3.1 Identifying Factors

To determine how homes would be heated throughout a day we need to consider all the factors that go into heat transfer and insulating a house. Some potential factors for us to consider are:

1. The ground area of the building
2. Amount of shade
3. The number of floors in the building
4. Number of people inside the housing unit
5. Sunlight throughout the day

1.3.2 Modelling

We intend to create our model using variations on numerous thermodynamic equations. These model how differently sized and insulated homes react to extreme heat. We will be looking at many methods of heat transfer, including radiation, convection and conduction. Using these, we intended to accurately model the flow of heat within our system.

We decided to model the system as having four components, the air inside the house, the air outside the house, the walls of the house and the ground beneath the house, each with their own respective temperature values. We can then model the heat exchanges between these components to work out an estimate for the temperature of the air inside the house.

We chose to use this model because it allows us to find the temperature of each component with relatively little effort, and work out how they influence each other.

We will also be factoring in how many people live in the building and the direct sunlight the building receives.

We chose these factors because we decided they were some of the most influential things that effected indoor temperature.

We came up with the following differential equation for the change in total energy in the air of the house, choosing to use differential equations because it would introduce a cause-effect delay in heating different components of our model, realistic to real life.

$$\dot{Q}_{Total} = \dot{Q}_{conduction-floor} + \dot{Q}_{radiation-people} + \dot{Q}_{conduction-walls} \tag{1}$$

This simply shows how the total change in heat energy is equal to the sum of the transfers of heat energy to and from different sources.

Using Fourier's law [5] we can find

$$\dot{Q}_{conduction-walls} = -k_I A \frac{dT}{dx} \tag{2}$$

Where:

- k_I = the thermal conductivity of the material, in this case insulation.
- A = the area of the walls and ceiling

$$\sqrt{floorarea} \cdot 4 \cdot n_{floors} + floorarea$$

- $$\frac{dT}{dx} = \frac{T_{house} - T_{walls}}{x_{grad}}$$

  where we take x_{grad} as the width of the insulation.

To find the heat produced per person we multiply the average surface area of a person [6] (calculated by summing the average surface area of men and that of women and dividing by 2) by the number of watts produced per m^2 of human skin [7], calculated using the sedentary activity number.

This gives us

$$1.75m^2 \cdot 58.2 \cdot 1.2 = 122.22W.$$

$$\dot{Q}_{radiation-people} = 122.22\,\mathrm{W} \cdot n_{people} \tag{3}$$

Where:

- n_{people} is the number of people in the house.

Furthermore, we can once again use Fourier's law [5] to get an equation for the rate of heat energy transfer from the ground (will be negative as heat is lost through the ground):

$$\dot{Q}_{conduction-floor} = k_F A \frac{dT}{dx} \tag{4}$$

Where:

- k_F is the thermal conductivity of the floor of the house.
- $A = floorarea.$
- $$\frac{dT}{dx} = \frac{T_{soil} - T_{house}}{x_{grad}}$$
  where we take x_{grad} as $5m$. We also take soil temperature to be a constant $12°C$ [4].

We also have an equation for the rate of change of the temperature of the walls:

$$\frac{dT_{walls}}{dt} = \frac{\dot{Q}_{radiation-sun} + \dot{Q}_{convection-outsideair} - \dot{Q}_{conduction-walls}}{m_{walls} \cdot c_{wall}} \tag{5}$$

Where:

- m_{walls} is the mass of the walls of the house based on the number of floors and $floorarea$, approximated from [8].
- c_{wall} is the specific heat capacity of the walls.

And

$$\dot{Q}_{convection-outsideair} = hA(T_{outside-air} - T_{walls}) \tag{6}$$

Where:

- $\dot{Q}_{convection-outsideair}$ = Rate of heat transfer from outside air to walls.
- A = Exposed surface area of house.
- ΔT is the difference in temperature between air and house.
- $h = 19.89$, the Convection Coefficient between air and red brick [9].

And furthermore

$$m_{walls} = (610 + 366 n_{floors}) \cdot floorarea \tag{7}$$

Where:

- n_{floors} = number of floors in the house.
- Constants represent pounds per floor per foot squared, then converted to SI units [8].

To calculate $Q_{radiation-sun}$ we can multiply the projected surface area of the house against the sun by the intensity of the sunlight and then again by the emissivity of the material of the house.

Intensity of sunlight on the ground is $1000W/m^2$ [10] and the emissivity of red brick is 0.93 [11].

To calculate the surface area projection of the house on the sun we can work out the individual projections of the planes making up the house and sum the ones that face the sun.

We have

$$Q_{radiation-sun} = A' \cdot I_0 \cdot s \tag{8}$$

Where:

- I_0 = intensity of sunlight on the ground ($1000W/m^2$ [10]).
- s = shade coefficient (percentage of intensity of I_0 reaching the surface throughout the day).

$$A' = \sum_{i=0}^{6} \begin{cases} A_i(\hat{n} \cdot \hat{s}), & \text{if the sun is above the horizon} \\ & \quad \text{and } A_i(\hat{n} \cdot \hat{s}) > 0 \\ 0, & \text{otherwise} \end{cases} \tag{9}$$

Where:

- A' = The projected area of the house by the sun.
- A_i = the area of face i:

$$A_i = \begin{cases} floorarea, & \text{if it is the ceiling} \\ \sqrt{floorarea} \cdot n_{floors} \cdot storeyheight, & \text{otherwise} \end{cases}$$

- $\hat{n}$ = The unit normal vector of the wall facing outside of the house.
- $\hat{s}$ = The unit vector towards the sun.

And if this is positive then we add it to our total area.

To calculate the unit vector pointing towards the sun we used the following equations [12].

$$\hat{s} = \begin{bmatrix} \cos\beta \\ \sin\beta \\ \sin\alpha \end{bmatrix} \tag{10}$$

$$\alpha = \arcsin(\sin\delta \sin\phi + \cos\delta \cos\phi \cos\gamma) \tag{11}$$

$$\beta = \begin{cases} \arccos\left(\frac{\sin\delta \sin\phi - \cos\delta \sin\phi \cos\gamma}{\cos\alpha}\right), & \gamma < 0 \\ 360^\circ - \arccos\left(\frac{\sin\delta \cos\phi - \cos\delta \sin\phi \cos\gamma}{\cos\alpha}\right), & \gamma \geq 0 \end{cases} \tag{12}$$

$$\delta = -23.45 \cdot \cos\left(\frac{360}{365}(d + 10)\right) \tag{13}$$

$$\gamma = 15^\circ \cdot (T - 12) \tag{14}$$

Where:

- ϕ is the latitude.
- d is the day of year.
- T is the time of day in hours.

Finally, to convert heat energy in a house to temperature we use the specific heat capacity of air and mass of air with the equation

$$\dot{Q}_{Total} = m_{air} c \Delta T \tag{15}$$

Where:

- $\dot{Q}_{Total}$ = Heat energy.
- m_{air} = mass of air.
- c = Specific heat capacity ($1005 JKg^{-1}C^{-1}$) [13].
- ΔT = Change in temperature.

This means we need to calculate the mass of air in the house. Assuming walls have negligible thickness and houses have a square base, we can find

$$m_{air} = floorarea \cdot \rho \cdot storeyheight \cdot n_{floors} \tag{16}$$

Where:

- $\rho = 1.293 Kg/m^3$ [14] (density of air).
- $storeyheight = 2.4m$ [15].

1.4 Results

Applying the model to Home 1 in Birmingham, UK [16], we get the input values

- $n_{floors} = 2$.
- $floorarea = 192 m^2$.
- $n_{people} = 4$.
- $s = 0.9$.
- $T_{outside-air}$ is taken to follow the data provided for temperature in Birmingham during a heatwave [16].
- Initial insider air temperature and wall temperature were taken to be the ambient temperature of the air at the beginning of the day.

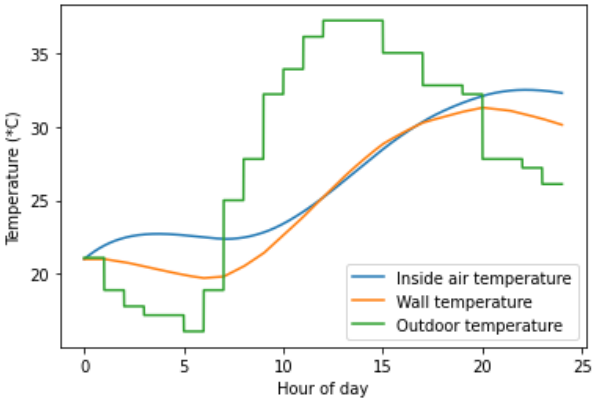


Figure 1: Temperatures for house 1

We then apply the same process for the remaining three houses.

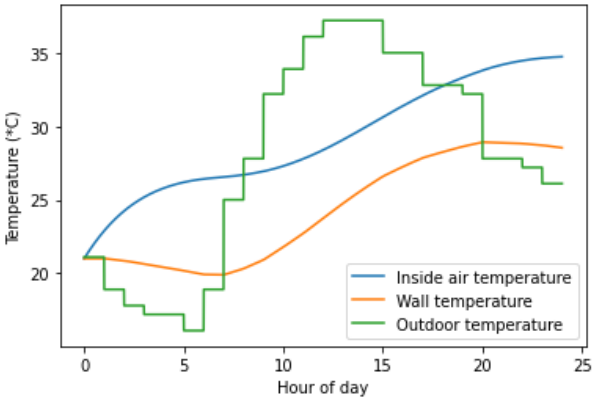


Figure 2: Temperatures for house 2

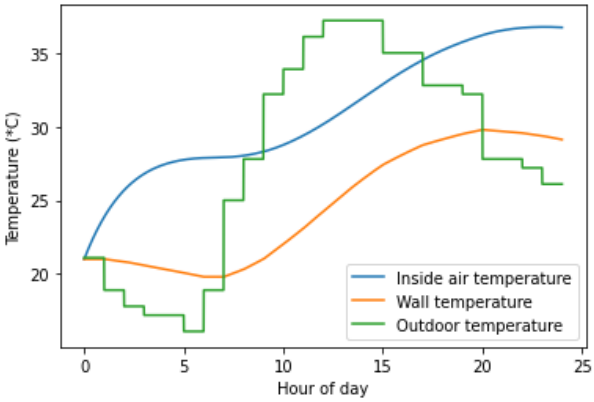


Figure 3: Temperatures for house 3

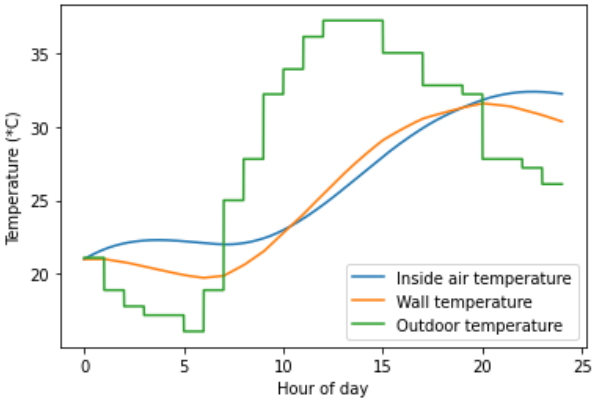


Figure 4: Temperatures for house 4

1.5 Strengths and Weaknesses

Our model is extremely physics based, with no reliance on using data of indoor house temperatures to base results on. This leaves the model vulnerable to being inaccurate if even weak factors are not taken into account for the model.

The model is also a large simplification of the real physics, making many assumptions that are likely to add up to having a substantial error.

By using differential equations for our model, we have allowed the indoor temperature to lag behind the outdoor temperature, which is realistic as it takes time for the outside conditions to effect inside ones.

By using a physics based model, we were able to integrate factors quite easily, allowing us to use things such as radiant body heat and energy lost through the floor. We are also able to account for changing external temperature and radiation from sunlight.

2 Question 2: Power Hungry

2.1 Restatement of the problem

We are tasked with developing a model that predicts Birmingham's peak power demands in summer months to help it avoid power outages due to high usage of air conditioning.

2.2 Hypothesis

We hypothesise that there is a positive correlation between the temperature in $^{\circ}C$ of Birmingham and the energy usage in kilowatt hours of Birmingham residents during the summer months.

2.3 Model Development

After finding data for peak temperature of a given day and half hourly data consumption across the course of a day we can filter for peak energy demands on each day and attempt to find a relation between the two sets [17, 18].

We first assumed that the two variables were linked linearly so used a linear regression model to calculate possible values for the equations constants.

$$\mu_i = \beta_0 + \beta_1 y_i + \beta_2 t_i \tag{17}$$

Where y represents the current year and t represents the temperature in $^\circ C$.

This seemed to produce consistent values, however, we weren't confident that this was the best relation so theorised that we are slowing in our improvement in technology to a point where we couldn't get a smaller amount of increased energy demand for a set temperature.

We then proceeded to attempt a logarithmic relationship.

$$\mu_i = \beta_0 + \beta_1 \log(y_i) + \beta_2 t_i \tag{18}$$

Unfortunately, this produced incredibly inconsistent datasets so we have ignored these results when considering our final model.

Justifications

We are treating this as a normal distribution, as it is a natural process with many different, complicated, seemingly random factors, that cumulates to a normal.

Also, it doesn't seem to have any particular reason or evidence that it can be represented by any other distribution, so trying to model it with a Normal distribution is our best bet.

We are claiming there is a linear relationship between heat and energy consumption.

Our reasoning was that it is the simplest relationship to demonstrate, so if it fits our data well, there would be no reason to reject it.

We also observed that it is the case if you look at any single given year, the relationship appears to be linear.

Furthermore, as shown in part 1, most heat transfer processes scale linearly with temperature. All of this together heavily implies that a linear relationship would fit our data.

We are also claiming that there is a linear relationship between the year and the energy consumption.

This seems counter-intuitive at first, as it would imply we will have over 100% energy efficiency at some point in the future, which as far as we understand is impossible; however, in the short term this holds true if you look at Figure 10.

2.3.1 Identifying factors

During the Summer months, we predict that the times of peak energy consumption will be during heat waves as the greatest increase in energy use will be from air conditioning, as such our factors will be the greatest temperature recorded on any given day as well as the greatest energy consumption on any given day [18, 17].

If this proves to be the case then existing models of increasing temperature can be used to predict rising peaks in energy demands.

2.4 Results

After putting our data through a MCMC algorithm, JAGS, with $x \sim N(\mu, \sigma^2)$, $\mu = \beta_0 + \beta_1 y + \beta_2 t$, and adjusting values (mostly widening the prior of β_0) we got a very nicely fitting model:

	Mean	SD	Naive SE	Time-series SE
β_0	1581769.43	26476.358	374.4322	381.82333
β_1	-770.00	13.153	0.1860	0.18969
β_2	26.88	18.824	0.2662	0.26175
Deviance	27029.13	2.927	0.0414	0.04139
Sigma	2349.67	43.281	0.6121	0.61299

Table 1: Summary statistics for parameter estimates

Looking at our values of β_1, β_2, and σ across our two cases for SE, Naive and Time-Series, we can see extremely low results.

This indicates that our parameter estimates are stable and have low uncertainty, which is a positive sign.

The values for β_0 seem quite high at 374 and 382, but given that it has a mean in the millions, this is actually quite low, and not something we should worry about.

	Point Est.	Upper C.I.
β_0	1	1.00
β_1	1	1.00
β_2	1	1.00
Deviance	1	1.00
Sigma	1	1.00

Table 2: Table of Estimates and Confidence Intervals

Our confidence intervals are also all 1 to the nearest 2dp, this is yet another excellent sign, especially given our wide priors, as it once again indicates an extremely low uncertainty.

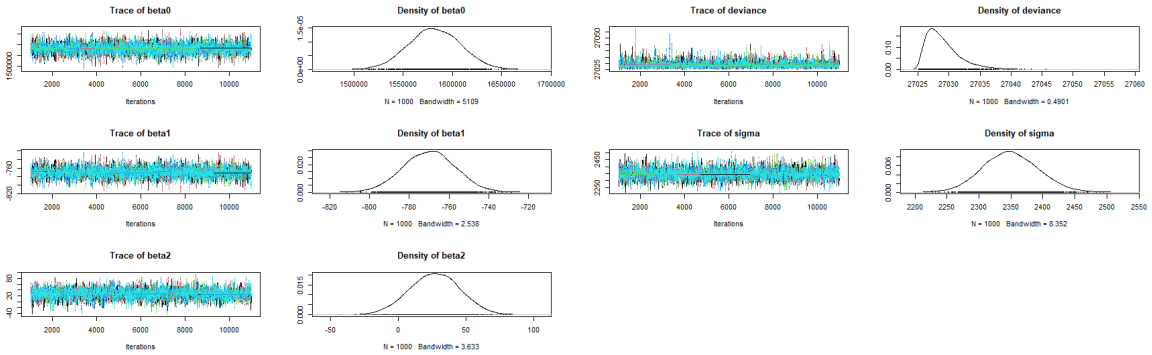


Figure 5: Trace and density plots for each variable

Nothing seems off about our traces, other than possibly deviance, but here deviance is not our focus, so this shouldn't affect our model.

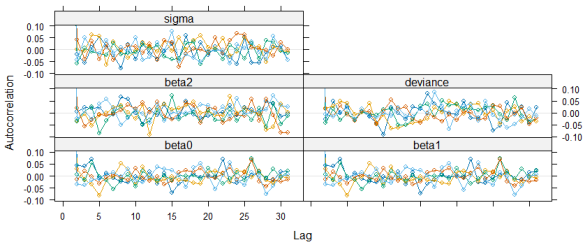


Figure 6: Autocorrelation plots for each variable

There are no visible trends or anomalies in our auto-correlation plots, so there are no issues that need addressing here.

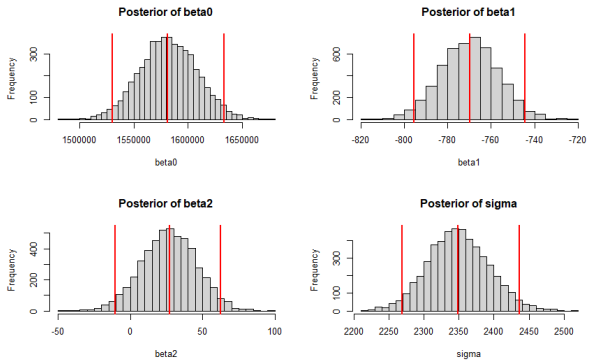


Figure 7: Posteriors with 2.5%, 50% and 97.5% marked

All results here are good, other than β_2 having a slight overlap with 0, this is quite concerning, as it implies that there is a chance within our critical values that it has the opposite relation to what we're predicting, but again, this may not be a significant problem, depending on the probability of this (this produces a probability not a likelihood due to using Bayesian techniques).

Looking closer at this result:

```
> prob_beta2_gt_0 <- mean(posterior_samples[, "beta2"] > 0)
> print(prob_beta2_gt_0)
[1] 0.9222
```

Figure 8: Chance of our β_2 value being positive given the data

Our result of 92.2% doesn't perfectly match our expectations but is still quite confident in such.

Looking also at the value for β_1:

```
> prob_beta1_lt_0 <- mean(posterior_samples[, "beta1"] < 0)
> print(prob_beta1_lt_0)
[1] 1
```

Figure 9: Probability of our β_1 being less than 0 given the data

This is a good result to see, as it confirms our expectations of a negative correlation.

One final check, does it agree with the data?

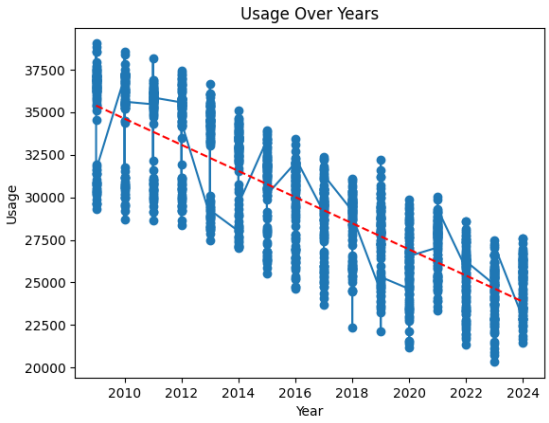


Figure 10: A linear regression through year against peak demand

According to this regression, our value for β_1 should be roughly -769, which is incredibly close to our found value of -770.

For the final visual check, let's plot the plane made by our means over our data, to ensure there is nothing obviously wrong with it.

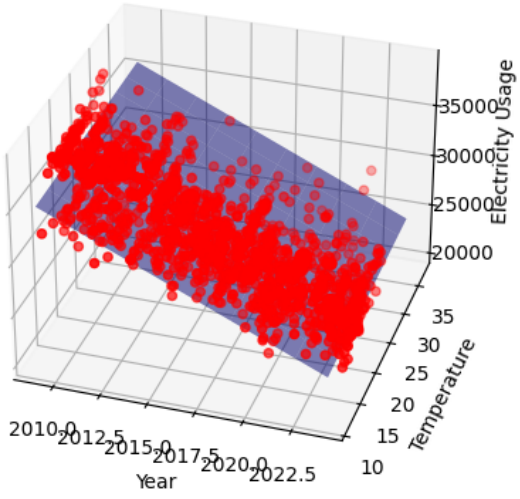


Figure 11: 3D plot produced by the plane made from our means over our data

It can be seen that our plane fits almost exactly with the original data, as it neatly cuts through our data through the middle, in the closest way possible, so this model does seem to work quite well, and can be used as our model for peak power output given a temperature and a year.

But we want to predict the output given the year alone, so we need to find a model for peak temperature given a year.

We received some data from the data sources given on exactly this [16], so let's analyse the trends of peak temperature over the last 15 years.

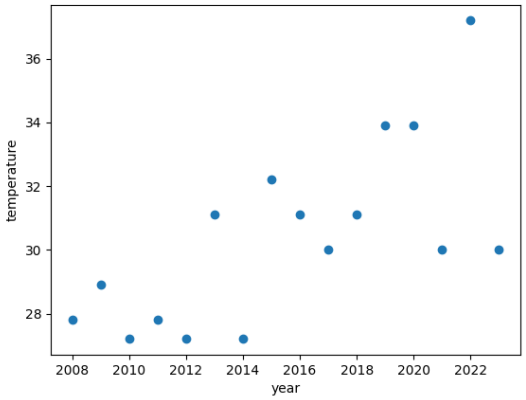


Figure 12: Peak temperature per year in Birmingham since 2008

This appears to form a linear relationship so we will first attempt to plot a straight line through the data, and see how well these fit.

Once again we are going to assume these follow a normal distribution, as climate models are complicated, and involve many factors with effects both positive and negative, that we aren't modelling, so a normal fits this best.

We can justify a linear relationship by looking at the graph, and seeing that the graph is not non-linear, so a linear is the best choice.

	Point Estimate	Upper C.I.
β_0	1.00	1.01
β_1	1.00	1.01
Deviance	1.00	1.00
Sigma	1.01	1.03

Table 3: Point Estimates and Upper Confidence Intervals

Once again, confidence intervals near 1, though further away than ideal, given the number of data points it's not a bad estimate, so there is no reason to discard the model over this, and we proceed forward.

	Mean	SD	Naive SE	Time-series SE
β_0	-757.2161	243.9706	4.4543	4.3493
β_1	0.3908	0.1211	0.0022	0.0022
Deviance	70.9618	2.5992	0.0475	0.0467
Sigma	2.2570	0.4564	0.0083	0.0085

Table 4: Posterior Mean, Standard Deviation, and Standard Errors

The value of β_1 being positive is a good sign, as that is what we expect.

Additionally the standard errors are all low enough to be considered acceptable, giving us good signs as to the success of fitting the model.

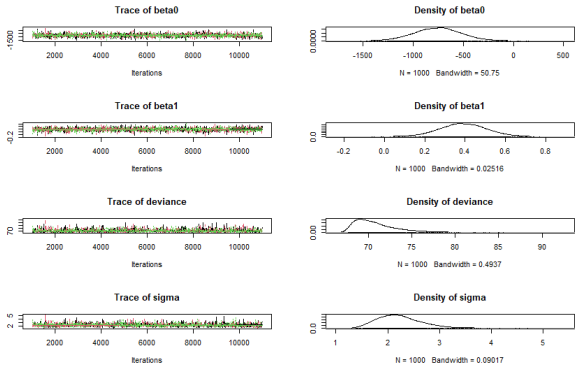


Figure 13: Trace and density plots for our variables

Once again, all the traces here are consistent with what we're expecting, and the density plots of our variables are all nicely spread so there is nothing anomalous to check here.

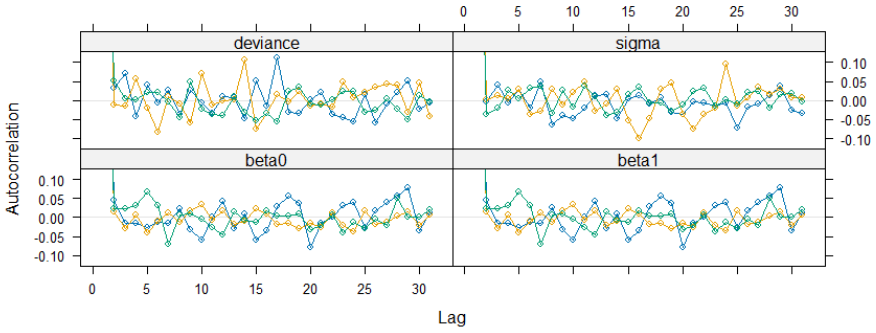


Figure 14: Autocorrelation plots for the MCMC fitting

The autocorrelation plots don't seem to contain any trends or gatherings, so there aren't any issues here, allowing us to move on.

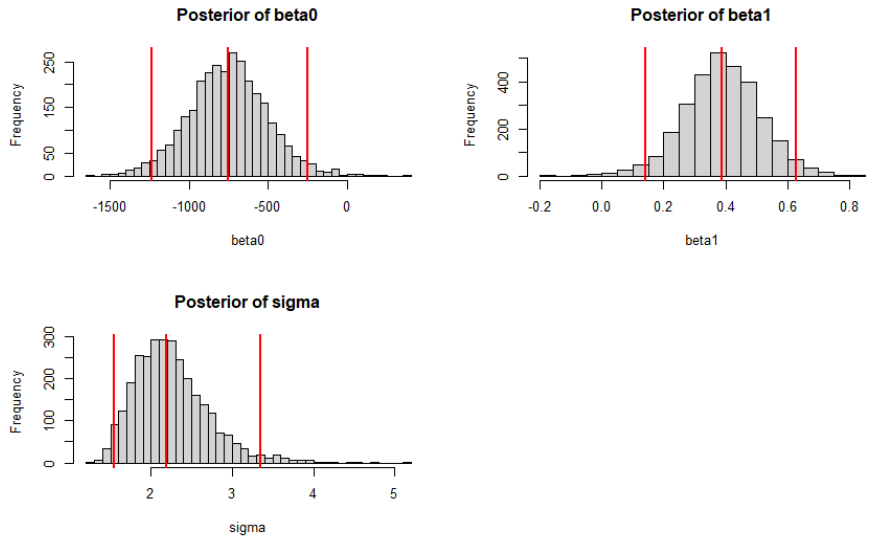


Figure 15: Posteriors of our variables

This model is consistent with the data and demonstrates that the climate is changing in order to increase the peak temperatures, which is to be expected.

Now that we have our completed models we can apply them to the question, Birmingham is responsible for 1.06% of the UK's power consumption, so we shall multiply our result by 0.0106 to get the amount of it which represents Birmingham (source: [19]).

We now have a model for what we predict the peak energy demands to be going forward from predicting the increase in peak temperatures during the summer months.

Taking a 5% lower and upper bound we see that over the next 20 years there is a 5% chance that the power demand of Birmingham exceeds 289kW and a 5% chance that the power demand never raises above 231kW.

Given our medians, we expect the peak temperature to be 33.6, as such the peak demand will be 248kW, which fits well with our expectations, so we do not expect any increases.

In fact, if we see what the bounds are for 2045, we get at 5% 45.6kW and at 95% 129kW, which means we are expecting a sharp decrease in demand, despite the increase in temperatures, as the current mean is not close to the bounds we predict.

2.5 Strengths and weaknesses

A major weakness with this model is the use of a linear prediction for power consumption does yield negative results in the far future, which is clearly wrong, so it must only work in shorter term predictions.

The model also does not take into account changes in climate action, and whether climate change might slow, or more pessimistically rise. Nor any other possible changes in trends.

One strength of the model is that we can find the aforementioned critical values, which is not only extremely useful for predictions, but also makes it quite easy to test the model in the not too far future.

One weakness is that it only accounts for the connection between year and peak temperature with the demand, whereas it may have been beneficial to instead use more variables.

However this low variable approach does have benefits. The model is closely fit to the data it was given, and no compromises had to be made.

3 Question 3: Beat the Heat

3.1 Restatement of the problem

This question asks us to analyse the various differences between Birmingham neighbourhoods in how they could respond to a power outage during a heat wave.

From this we should define vulnerability scores for each neighbourhood and using these advise the city on how best to allocate resources during such an emergency situation.

We should justify all the factors that we consider when assigning these scores.

3.2 Factors to consider

The data we have been provided gives us information about the population demographics of Birmingham - this means we have the option to consider any combination of the following factors:

- Types of home
- Methods of travel
- Age of inhabitants
- Average income
- Over crowdedness
- Unemployment
- Average level of Education
- Number of people claiming benefits
- Environmental score [20]
- Access to/ownership of vehicles

Of these, the most likely factors to affect a local authority's response would be helping those who are unable to help themselves - in this case, the elderly and young children.

3.3 Model Development

We came up with two possible models to consider how vulnerable a neighbourhood might be, one looks at how many people in those constituencies suffer from each negative factor and our second looks at the Spearman rank correlation of neighbourhoods containing each factor.

3.3.1 Chance of possessing factors

Using the data provided to us by the M3C team, we can calculate the probability that an individual in each Birmingham constituency is classified under any one of our variables.

For example, the chance of someone from Ladywood being over 65 years old is

$$8000/126324 = 0.0633.$$

By multiplying multiple of these values together we can find the amount of people in each constituency who are included in these sets, by adding weight to each factor based on how detrimental we think it would be to managing a power cut during a heat wave and summing across every permutation, we can find the vulnerability score for that neighbourhood.

3.4 Calculating Vulnerability score

To calculate vulnerability score we decided to work out the numbers of people in different constituencies with different numbers of vulnerabilities.

Looking through data provided, we identified vulnerabilities to be:

1. High proportion of over 65s.

   Justification: Old people are much more likely to die from extreme heat [21].

2. High percentage of overcrowded households.

   Justification: Overcrowded houses will heat up more $Q_{radiation-people}$.

3. Low percentage of households with 2+ cars or vans.

   Justification: People with no cars can't go to parks and cool down [2].

Once we had identified vulnerabilities we could start working out probabilities for different combinations of them.

By converting all of the data into percentages of the total population in the constituency and assuming each factor to be independent, we could work out the probability of that specific combination occurring.

Starting at a chance of 1, for each factor, if we were considering it in the combination then we multiplied the chance by the probability of that factor and otherwise we multiplied by 1-the probability of it occurring.

Once we had the probabilities of each combination occurring, we can multiply by the population of the constituency to work out the number of people with each combination of the vulnerabilities.

By then multiplying this by a weight of the number of vulnerabilities and summing them for each combination, we effectively calculate a vulnerability score for each constituency.

Using Python, we came up with the numbers below.

Constituency	Vulnerability score
Birmingham Edgbaston	74635.22
Birmingham Erdington	86383.37
Birmingham Hall Green and Moseley	93256.79
Birmingham Hodge Hill and Solihull North	71007.65
Birmingham Ladywood	66209.88
Birmingham Northfield	81194.66
Birmingham Perry Barr	78068.43
Birmingham Selly Oak	79728.78
Birmingham Yardley	82795.98
Sutton Coldfield	84436.60

Table 5: Vulnerability scores by constituency

3.5 Using vulnerability score for optimal housing construction

Using our vulnerability scores, we can rank the constituencies in order of need.

In those of high need, developing more heat resistant housing would be greatly beneficial in reducing risk for those most vulnerable, providing shelter from the scorching heat.

For the most effective housing to help combat overcrowded houses in vulnerable areas, we can use our model from question 1 to determine the building with the best heat resistance, which would keep it the coolest.

If we combine this with green roofs, which can reduce nearby air temperatures by up to almost $10°C$ [3], we can increase green spaces without having to sacrifice floorspace needed to reduce overcrowded houses.

The most effective building at keeping the heat low was a 1 bedroom apartment as it peaks at the lowest temperature.

Building apartment blocks with few floors and green roofs is a good solution to both problems, and should help reduce risks from extreme temperatures.

3.6 Strengths and weaknesses

One weakness of our model is its failure to consider different types of housing; as shown in our answer to the first question, the type of house you live in has a massive impact on how long it takes to reach peak temperature during a heat wave without air-conditioning.

Consequently, a neighbourhood that suffers from many problems but which has lots of well insulated small houses will be given a high vulnerability score despite it being able to survive far longer before reaching its peak temperature, and will have a peak temperature that is far lower than other constituencies.

A powerful strength of our model is adaptability, it can grow with any factor that can be considered as a percentage of the population.

Meaning, different variables can be proposed and substituted in without any change to the core of the model in order to see how different factors affect vulnerability scores.

4 Conclusion

4.1 Further studies

4.1.1 Model one: Hot to Go

In order to further improve upon this model we would add more variation through the use of modifications like adding sized windows and different glazings to make more complicated homes.

Changes like this would allow the model to be customizable to any city who wished to use it for their typical homes, making it more applicable to all situations while also making it more specific for each job.

Our model also requires the house to not have any air conditioning, extra features like internal cooling could make this model applicable to day to day situations not just to emergency cases like the ones discussed in this report.

4.1.2 Model two: Power Hungry

This model could be further improved by simply adding more data, though we struggled to find other sources.

Also considering different relationships which we didn't have time to test, for example

$$\log \mu_i = \beta_0 + \beta_1 y_i + \beta_2 t_i. \tag{19}$$

4.1.3 Model three: Beat the Heat

To improve this model further, we could find a way to include the environment rating in our vulnerability score, as green spaces can reduce temperatures by up to $4°C$ [2], so areas without enough green space will be even hotter.

4.2 Summary

Our work has produced three models that can be used for predicting the responses of certain systems within Birmingham during a heatwave.

Our first uses thermodynamic equations to model the increase in internal temperature of various homes throughout hot days in order to see which style of buildings best resist heating and have a minimum peak temperature.

Our second finds relations between highest temperature recorded and peak energy demand in a day to try predict the future demands on Birmingham's power grid with the effects of climate change.

This will prevent black outs during heat waves which could cause serious harm to infrastructure and loss of life, especially among the elderly and most vulnerable areas.

Our third model works to see how different parts of Birmingham would suffer from a power outage, calculating vulnerability scores for each neighbourhood and designing an equation which can be easily edited with the factors of interest at the time.

We then follow this up with some recommendations on how vulnerable communities can be changed to make them more robust and resistant to heat waves.

5 References

[1] Fire protection association. *Industry guide for living walls as cladding released*. URL: https://www.thefpa.co.uk/news/industry-guide-for-living-walls-as-cladding-released.

[2] *Learn about Heat Island Effects*. URL: https://www.epa.gov/heatislands/learn-about-heat-island-effects. Date Accessed - 01/03/2025.

[3] US environmental protection agency. *Using Green Roofs to Reduce Heat Islands*. URL: https://www.epa.gov/heatislands/using-green-roofs-reduce-heat-islands.

[4] British Geological Survey. *Temperature and Thermal Properties (basic)*. URL: https://shop.bgs.ac.uk/resources/shop/doc/example/product/modules/C012.pdf.

[5] Wikipedia. *Thermal Conduction*. URL: https://en.wikipedia.org/wiki/Thermal_conduction. Date Accessed: 28/02/2025.

[6] MD Melissa Conrad Stöppler. "Definition of body surface area". In: *None* (1987). URL: https://www.rxlist.com/body_surface_area/definition.htm#:~:text=Average%20body%20surface%20area%20for,10%20years)%3A%201.14%20m2.

[7] Cornell University Ergonomics Web. "DEA3500: Ambient Environment: Thermal Conditions". In: *None* (2025). URL: https://ergo.human.cornell.edu/studentdownloads/DEA3500notes/Thermal/thcondnotes.html#:~:text=Every%20square%20foot%20of%20body,of%20about%2019%20matches%2Fhour.&text=Du%20Bois%20area%3A%20The%20surface,%3D%20356%20Btu's%20per%20hour)..

[8] Benjamin Ehinger. *How Much Does a House Weigh? Understanding the Averages and Variables*. URL: https://wasteremovalusa.com/blog/how-much-does-a-house-weigh/#:~:text=On%20a%20broad%20scale%2C%20house,building%20materials%20and%20construction%20methods.. Date Accessed: 28/02/2025.

[9] Rafael Almanza Jesús Chávez-Galán and Neftalí Rodríguez Cuevas. *convective heat transfer coefficients*. URL: https://www.scielo.org.mx/pdf/ccid/v5n2/v5n2a3.pdf.

[10] Woods Hole Oceanographic Institution. *Solar Radiation*. URL: https://www.whoi.edu/science/AOPE/mvco/description/SolRad.html.

[11] Omega. *Emissitivity of common materials*. URL: https://www.omega.co.uk/literature/transactions/volume1/emissivityb.html.

[12] PhD Wojciech Sas. *Computing the Sun's location*. URL: https://www.omnicalculator.com/physics/sun-angle.

[13] Virginia Institute of Marine Science Christopher J. Petrone Virginia Sea Grant. *Specific heat capacity of air*. URL: https://masweb.vims.edu/bridge/datatip.cfm?Bridge_Location=archive0909.html.

[14] Nasa Earth Data. *Air mass/density*. URL: https://www.earthdata.nasa.gov/topics/atmosphere/air-mass-density#:~:text=Pure%2C%20dry%20air%20has%20a,of%20at%20least%2050%20km.. Date Accessed: 28/02/25.

[15] Wall Panels World. *Standard Ceiling Height UK | New 2025 Regulations*. URL: https://wallpanelsworld.co.uk/blog/standard-ceiling-height/#:~:text=The%20typical%20ceiling%20height%20in,furniture%20space%2C%20and%20heating%20efficiency.. Date Accessed: 28/02/25.

[16] MathWorks Math Modeling Challenge 2025. *Hot Button Issue*. URL: https://m3challenge.siam.org/897bjhb54cgfc/. Curated data.

[17] Neso. *Historic Demand Data*. URL: https://www.neso.energy/data-portal/historic-demand-data/historic_demand_data_2009. Date Accessed: 28/02/2025.

[18] Hadley Server. *Met Office Hadley Centre Central England Temperature*. URL: https://hadleyserver.metoffice.gov.uk/hadcet/data/download.html. Date Accessed: 28/02/2025.

[19] Dukes. *Chapter 5: electricity*. URL: https://assets.publishing.service.gov.uk/media/66a7da1bce1fd0da7b592f0a/DUKES_2024_Chapter_5.pdf.

[20] Birmingham City Council. *Open Space Assesment 2022*. URL: https://www.birmingham.gov.uk/downloads/file/24587/open_space_assessment_2022. Date Accessed: 28/02/2025.

[21] Gov.uk. *Heat mortality monitoring report: 2022*. URL: https://www.gov.uk/government/publications/heat-mortality-monitoring-reports/heat-mortality-monitoring-report-2022.

6 Code Appendix

6.1 Question1.py

```
import math
from math import sin,cos
import matplotlib.pyplot as plt
tempdata = [
21.1,
18.9,
17.8,
17.2,
17.2,
16.1,
18.9,
25.0,
27.8,
32.2,
33.9,
36.1,
37.2,
37.2,
37.2,
35.0,
35.0,
32.8,
32.8,
32.2,
27.8,
27.8,
27.2,
26.1
]
def calculatesun(T,d,lat):
    gamma = 15 * (T-12)
    delta = -23.45 * math.cos(math.radians(360*(d+10)/365))
    alpha = math.asin(sin(delta)*sin(lat)+cos(delta)*cos(lat)*cos(gamma))

    delta = math.radians(delta)
    alpha = math.radians(alpha)
    gamma = math.radians(gamma)

    beta = math.acos((sin(delta)*cos(lat)-cos(delta)*sin(lat)*cos(gamma))/
    cos(alpha))
    if gamma >= 0:
        beta = 360 - beta


    return [cos(beta),sin(beta),sin(alpha)]

def getenergy(second):
    hour = second / (60**2)
    day = 197
    lat = 52.49
    intensity = 1000
    emm = 0.93

    normals = [
    [0,0,1],
```

```
    [0,1,0],
    [1,0,0],
    [-1,0,0],
    [0,-1,0],
    [0,0,-1]
    ]
    sun = calculatesun(hour, day, 52.49)
    sa = [10,10,10,10,10,10]
    area = 0
    for i,item in enumerate(normals):
        p =(sa[i] * sum([x*y for x,y in zip(sun,item)]))
        if p >0:
            area += p

    energy = intensity * area * emm #* #sin(3.1415 * second/(24*60**2) )
    if 5 + 4/60<hour<21 + 21/60:
        return energy
    else:
        return 0



def get(time,walltemp,temp,outdoor):
    shade = 1

    outside_temp = outdoor
    people = 3
    k = 0.3
    floors = 3

    floorarea = 139*1
    insulation_width = 0.6
    A = floorarea**0.5 * 4 * floors + floorarea
    wall_mass =   (125 + 75*floors) * floorarea * 4.88243
    wall_shc = 840
    air_shc = 1000
    air_mass = floorarea * 1.293 * 2.4*floors

    q_people = 122.22 * people
    #print("q_")
    q_convection = (walltemp - outside_temp) * A * 19.89#h_c/c_p


    delta_wall_temp = (getenergy(time)*shade-q_convection) / (wall_mass *
    wall_shc)

    walltemp += delta_wall_temp - k * A * (walltemp-temp) / (
    insulation_width * wall_mass * 840)
    q_conduction = - 0.3 * floorarea * (temp - 12)/5
    q_conduction_walls = k * A * (walltemp-temp) / insulation_width
    q_t = q_conduction_walls  + q_conduction + q_people

    temp += q_t / (air_mass * air_shc)

    return temp,walltemp,[getenergy(time),q_conduction,q_conduction_walls]

time = 0
walltemp = 21
temp = 21
xpoints = []
```

```
ypoints = []
y2points = []
y3points = []
xpoints2 = []
for i in range(24 * 60**2):

    outdoor = tempdata[(i//(60**2))]
    temp,walltemp,info = get(i,walltemp,temp,outdoor)
    xpoints.append(i / 60**2)
    xpoints2.append(i //(60**2))
    ypoints.append(temp)
    y2points.append(walltemp)
    y3points.append(outdoor)


plt.plot(xpoints, ypoints,label="Inside air temperature")
plt.plot(xpoints, y2points,label="Wall temperature")
plt.plot(xpoints, y3points,label="Outdoor temperature")
plt.xlabel("Hour of day")
plt.ylabel("Temperature (*C)")
plt.legend()
plt.show()
```

6.2 vuln.py

```
#a  = [8.9, 6.1,29.4, 4.3,  20.16]

#old = [0.1461670480549199, 0.140524379024839, 0.12489196888703946,
    0.10952200962314458, 0.0633292169342326, 0.16522802601960332,
    0.10726686398374342, 0.1411638781902247, 0.12546245727775623,
    0.22947294419287473]

#p#op = 1000
populations =
    [108376,130440,124964,97889,126324,112838,113185,113981,113524,96726]
crowded = [8.9,8,13,11,28.3,8.2,12.9,8,11.6,3.7]
cars = [20.16,19.2,21.49,21.21,9.41,21.98,18.84,21.02,21.18,34.24]
old = [0.1461670480549199, 0.140524379024839, 0.12489196888703946,
    0.10952200962314458, 0.0633292169342326, 0.16522802601960332,
    0.10726686398374342, 0.1411638781902247, 0.12546245727775623,
    0.22947294419287473]
old = [o * 100 for o in old]
for p in range(10):
    a = [crowded[p],cars[p],old[p]]
    pop = populations[p]

    bins = []
    for i in range(2**len(a)):
        bins.append(bin(i)[2:])

    total = 0
    probs = []
    for item in bins:
        mult = 1
        for i in range(len(a)-1):
            try:
                if item[i] == "1":
                    mult *= a[-i+1]  / 100
```

```
                else:
                    mult *= 1 - a[-i+1]  / 100
            except:
                mult *= 1 - a[-i+1]  / 100

        probs.append(mult)

        total += mult

    vul = 0


    for i,item in enumerate(probs):
        probs[i] = item/total


    for i,item in enumerate(bins):
        t = 0
        for item2 in item:
            if item2 == "1":
                t += 1
        vul += pop * t * probs[i]


    print(vul)
```

6.3 Question2-model1.R

```
library(coda)
library(R2jags)

plot_posterior <- function(samples, param_name) {
  hist(samples, breaks = 30, main = paste("Posterior of", param_name), xlab
     = param_name)
  abline(v = quantile(samples, c(0.025, 0.5, 0.975)), col = "red", lwd = 2)
}

compute_quantiles <- function(samples, probs = c(0.025, 0.5, 0.975)) {
  apply(samples, 2, quantile, probs = probs)
}

modelString = "
model{
   for( i in 1 : n ) {
      x[i] ~ dnorm(mu[i], tau)
      mu[i] <- beta0 + beta1 * y[i] + beta2 * t[i]
   }
   beta0 ~ dnorm(0.0, 1.0E-99)
   beta1 ~ dnorm(0.0, 1.0E-5)
   beta2 ~ dnorm(0.0, 1.0E-5)
   tau ~ dgamma(0.001, 0.001)
   sigma <- 1 / sqrt(tau)
}
"

.temp = file("model.txt", "w") ; writeLines(modelString, con = .temp) ;
    close(.temp)
```

```r

data <- read.csv("yes.csv")
#data <- data.frame(x = x, y = y, t = t)
y <- data$y
x <- data$x
t <- data$t
dataList <- list(
  n = length(y),
  x = x,
  y = y,
  t = t
)

init1 <- list(beta0 = 0, beta1 = -10, beta2 = 10, tau = 1.0)
init2 <- list(beta0 = 0.5, beta1 = -5, beta2 =0, tau = 1.0)
init3 <- list(beta0 = 1, beta1 = -11, beta2 = 1, tau = 1.0)
init4 <- list(beta0 = 1.5, beta1 = -15, beta2 = 1.5, tau = 1.0)
init5 <- list(beta0 = 2, beta1 = -20, beta2 = 2, tau = 1.0)
inits <- list(init1, init2, init3, init4, init5)

params <- c("beta0", "beta1", "beta2", "sigma")

results <- jags(
  data = dataList,
  inits = inits,
  parameters.to.save = params,
  model.file = "model.txt",
  n.burnin = 1000,
  n.iter = 11000,
  n.thin = 10,
  n.chains = 5,
  DIC = TRUE,
  working.directory = getwd()
)

results_mcmc <- as.mcmc(results)

plot(results_mcmc)

gelman.diag(results_mcmc)

acfplot(results_mcmc)

summary(results_mcmc)

posterior_samples <- as.matrix(results_mcmc)

quantiles <- compute_quantiles(posterior_samples)

par(mfrow = c(2, 2))
plot_posterior(posterior_samples[, "beta0"], "beta0")
plot_posterior(posterior_samples[, "beta1"], "beta1")
plot_posterior(posterior_samples[, "beta2"], "beta2")
plot_posterior(posterior_samples[, "sigma"], "sigma")

print(quantiles)

file.remove("model.txt")
graphics.off()
```

6.4 Question2-model2.R

```
library(coda)
library(R2jags)

plot_posterior <- function(samples, param_name) {
  hist(samples, breaks = 30, main = paste("Posterior of", param_name), xlab
     = param_name)
  abline(v = quantile(samples, c(0.025, 0.5, 0.975)), col = "red", lwd = 2)
}

compute_quantiles <- function(samples, probs = c(0.025, 0.5, 0.975)) {
  apply(samples, 2, quantile, probs = probs)
}

modelString = "
model{
   for( i in 1 : n ) {
      t[i] ~ dnorm(mu[i], tau)
      mu[i] <- beta0 + beta1 * y[i]
   }
   beta0 ~ dnorm(0.0, 1.0E-)
   beta1 ~ dnorm(0.0, 1.0E-5)
   tau ~ dgamma(0.001, 0.001)
   sigma <- 1 / sqrt(tau)
}
"

.temp = file("model.txt", "w") ; writeLines(modelString, con = .temp) ;
    close(.temp)

data <- read.csv("yeah.csv")
#data <- data.frame(x = x, y = y, t = t)
y <- data$y
t <- data$t
dataList <- list(
  n = length(y),
  y = y,
  t = t
)

init1 <- list(beta0 = 0, beta1 = -10, tau = 1.0)
init2 <- list(beta0 = 0.5, beta1 = -5, tau = 1.0)
init3 <- list(beta0 = 1, beta1 = -11, tau = 1.0)
inits <- list(init1, init2, init3)

params <- c("beta0", "beta1" , "sigma")

results <- jags(
  data = dataList,
  inits = inits,
  parameters.to.save = params,
  model.file = "model.txt",
  n.burnin = 1000,
  n.iter = 11000,
  n.thin = 10,
  n.chains = 3,
  DIC = TRUE,
  working.directory = getwd()
)
```

```r

results_mcmc <- as.mcmc(results)

plot(results_mcmc)

gelman.diag(results_mcmc)

acfplot(results_mcmc)

summary(results_mcmc)

posterior_samples <- as.matrix(results_mcmc)

quantiles <- compute_quantiles(posterior_samples)

par(mfrow = c(2, 2))
plot_posterior(posterior_samples[, "beta0"], "beta0")
plot_posterior(posterior_samples[, "beta1"], "beta1")
plot_posterior(posterior_samples[, "beta2"], "beta2")
plot_posterior(posterior_samples[, "sigma"], "sigma")

print(quantiles)

file.remove("model.txt")
graphics.off()
```

6.5 critical-values.R

```r
years <- 2026:2045
threshold_range <- c(20000, 50000)
beta0_main <- posterior_samples_main[, "beta0"]
beta1_main <- posterior_samples_main[, "beta1"]
beta2_main <- posterior_samples_main[, "beta2"]
sigma_main <- posterior_samples_main[, "sigma"]

beta0_temps <- posterior_samples_temps[, "beta0"]
beta1_temps <- posterior_samples_temps[, "beta1"]
sigma_temps <- posterior_samples_temps[, "sigma"]

thresholds <- seq(threshold_range[1], threshold_range[2], length.out = 100)
probabilities <- numeric(length(thresholds))

for (j in seq_along(thresholds)) {
  threshold <- thresholds[j]
  prob_years <- numeric(length(years))

  for (i in seq_along(years)) {
    y_value <- years[i]

    t_samples <- beta0_temps + beta1_temps * y_value + rnorm(length(beta0_
   temps), 0, sigma_temps)

    x_samples <- beta0_main + beta1_main * y_value + beta2_main * t_samples
    + rnorm(length(beta0_main), 0, sigma_main)

    prob_years[i] <- mean(x_samples > threshold)
  }

```

```
  probabilities[j] <- 1 - prod(1 - prob_years)
}

threshold_0_05 <- thresholds[which.min(abs(probabilities - 0.05))]*0.0106
threshold_0_95 <- thresholds[which.min(abs(probabilities - 0.95))]*0.0106

year <- 2045
beta0_main <- posterior_samples_main[, "beta0"]
beta1_main <- posterior_samples_main[, "beta1"]
beta2_main <- posterior_samples_main[, "beta2"]
sigma_main <- posterior_samples_main[, "sigma"]

beta0_temps <- posterior_samples_temps[, "beta0"]
beta1_temps <- posterior_samples_temps[, "beta1"]
sigma_temps <- posterior_samples_temps[, "sigma"]

t_samples <- beta0_temps + beta1_temps * year + rnorm(length(beta0_temps),
    0, sigma_temps)

x_samples <- beta0_main + beta1_main * year + beta2_main * t_samples +
    rnorm(length(beta0_main), 0, sigma_main)

critical_value_0_05 <- quantile(x_samples, 0.05)
critical_value_0_95 <- quantile(x_samples, 0.95)

print(paste(critical_value_0_05*0.0106))
print(paste(critical_value_0_95*0.0106))
```